

My (Sarita Sapkota) Role Summary As First Author in the Submitted Manuscript of the Paper

Paper Title: *A Multi-Scale Deep Learning Approach for the Study of Hairline Cracking using Hybrid CNN-Transformer Model*

Affiliation: Department of Civil Engineering, Kathmandu University

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My Role as First Author

As the **first author** and a **civil engineering researcher**, I was primarily responsible for conceptualizing, validating, and interpreting this study from a **structural health monitoring and materials engineering perspective**. My contributions focused on ensuring that the technical design, data, and experimental outcomes were consistent with the **real-world behavior of cracks in civil structures** and aligned with **engineering diagnostic standards**.

1. Conceptualization and Research Design

I initiated the research concept by identifying the limitations of traditional crack detection methods used in **structural health assessment** and proposed the use of a **multi-scale hybrid deep learning model** to overcome these challenges.

My focus was to ensure that the research remained **grounded in the civil engineering context**, emphasizing the need for early detection of **hairline cracks** to support **preventive maintenance and durability assessments**. I also collaborated with co-authors to align the model's capabilities with real inspection needs and existing **field monitoring techniques**.

2. Literature Review

I conducted the **literature review** on conventional and modern **crack detection techniques** in civil infrastructures.

This included examining:

- **Traditional image-processing methods** such as Sobel, Canny, and thresholding, and their limitations under **variable lighting and surface textures**.
- **Civil-engineering-based studies** like **Cha et al. (2017)**, which applied deep learning for tunnel crack detection, and the **SDNET2018 dataset (Maguire et al., 2018)** used for benchmarking concrete crack analysis under real conditions.
- Prior research such as **Zou et al. (2018)**, highlighting the importance of annotated datasets for reliable crack identification.

3. Methodology and Experimental Setup (Civil Perspective)

I contributed to the **methodology design** by ensuring the experimental setup reflected **practical inspection environments**.

Specifically, I:

- Designed the **data acquisition strategy** for collecting field images from **concrete structures in Nepal**, covering diverse surface types and environmental conditions.
- Provided guidance on how to interpret **multi-scale feature learning** outputs in relation to **crack morphology, orientation, and propagation behavior**.
- Ensured that the hybrid model's design and its performance metrics (IoU, F1-Score, precision, recall) were analyzed not just statistically but also **from an engineering significance standpoint**, i.e., how effectively it could detect early-stage deterioration.

4. Model Validation through Civil Engineering Expertise

A major part of my role was in **model validation** and **cross-verification of results** from a **civil engineering viewpoint**.

I examined how accurately the model's segmentation outputs represented **real crack geometry** and **damage progression**.

Key aspects I focused on included:

- Verifying **crack boundary precision**, ensuring that predicted crack widths matched field measurements within acceptable tolerance levels.
- Evaluating the **consistency of detection** across variable surface textures, concrete finishes, and shadowing—conditions commonly faced in site inspections.
- Correlating the **temporal crack evolution analysis** (growth in length and width) with **load-induced behavior** observed in actual structural elements.

Through this process, I ensured that the **hybrid CNN-Transformer model's outputs were scientifically interpretable and practically applicable** in structural condition monitoring and **predictive maintenance frameworks**.

5. Result Analysis and Interpretation

I played a central role in interpreting the **quantitative and qualitative results**.

While the co-authors contributed to model optimization and coding, I analyzed how the performance metrics—particularly **IoU, F1-score, and recall**—translated to **field-level reliability** for civil structures.

I contextualized the outcomes in terms of:

- **Detection reliability for micro-cracks (<0.3 mm),**
- **Impact on long-term durability,** and
- **Potential for integrating the model into real-time inspection systems** such as drones or fixed monitoring cameras.

6. Drafting and Refinement

I drafted the **Introduction, Literature Review, Methodology (civil sections), and Discussion** parts of the paper.

In these sections, I emphasized the engineering significance of **multi-scale learning**, the need for **automated visual inspection**, and the **real-world interpretability** of the model's predictions.

I also contributed to refining the **Abstract** and **Conclusion**, ensuring that the study highlighted its value for **structural engineers** and **infrastructure asset managers**.

Summary of My Contributions

Category	My Specific Role
Conceptualization	Defined problem statement and research scope from a civil engineering standpoint.
Literature Review	Synthesized SHM and crack detection studies to identify knowledge gaps.
Methodology Design	Developed civil-based validation approach and field data protocols.
Data Collection	Led field image collection and ground truth validation.
Model Validation	Evaluated and verified outputs from an engineering accuracy perspective.
Result Interpretation	Connected model performance to real-world structural integrity assessment.
Drafting & Editing	Authored key civil-focused sections and finalized manuscript.

Conclusion

In this study, I served as the **bridge between civil engineering principles and AI-based modeling techniques**. My primary contribution lay in ensuring that the proposed hybrid CNN-Transformer model was not only computationally efficient but also **technically meaningful and field-validated** for **structural health monitoring applications**. Through my expertise in **material behavior, field inspection, and crack propagation analysis**, I ensured that the model's validation and interpretation adhered to the **real demands of civil infrastructure assessment**.